

Intermediate Code Generation & Optimization Report

Domain-Specific Language Compiler for Financial Computing

Target Expression evaluated:

$((\text{principal} * \text{rate} * \text{time}) / 100) + (\text{principal} * (1 + \text{rate} / 100) ^ \text{time}) - \text{fees}$

Optimization Reduction: 0.00% (From 9 to 9 Instructions)

1. Unoptimized Three-Address Code (TAC)

```
t1 = principal * rate
t2 = t1 * time
t3 = t2 / 100
t4 = rate / 100
t5 = 1 + t4
t6 = t5 ^ time
t7 = principal * t6
t8 = t3 + t7
t9 = t8 - fees
```

2. Quadruple Representation Table

(0)	*	principal	rate	t1
(1)	*	t1	time	t2
(2)	/	t2	100	t3
(3)	/	rate	100	t4
(4)	+	1	t4	t5
(5)	^	t5	time	t6
(6)	*	principal	t6	t7
(7)	+	t3	t7	t8
(8)	-	t8	fees	t9

3. Triple Representation Table

(0)	*	principal	rate
(1)	*	(0)	time
(2)	/	(1)	100
(3)	/	rate	100
(4)	+	1	(3)
(5)	^	(4)	time
(6)	*	principal	(5)
(7)	+	(2)	(6)
(8)	-	(7)	fees

4. Optimized Three-Address Code (Post CSE & DCE)

```
t1 = principal * rate
t2 = t1 * time
t3 = t2 / 100
t4 = rate / 100
t5 = 1 + t4
t6 = t5 ^ time
t7 = principal * t6
t8 = t3 + t7
t9 = t8 - fees
```